

1D Stochastic Z-Axis Physically Unclonable Functions

A Collaboration Whitepaper for Experimental Validation

Aaron Garcia

Independent Researcher

aaron@garcia.ltd

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No institutional affiliation. The author's claimed professional background (30 years, Systems Engineering) is stated for context but cannot be independently verified from this document alone. Readers are encouraged to evaluate the technical content on its merits.

1. Introduction

Counterfeit consumer goods remain a significant global problem. Existing authentication technologies—QR codes, holographic labels, RFID/NFC, taggant inks—each carry known limitations in security, cost, or readout infrastructure, though several (particularly RFID and DNA taggants) continue to perform adequately within their respective niches.

This whitepaper proposes a concept for a new class of physical unclonable function (PUF) that would encode its cryptographic entropy in the z-axis—the depth dimension of a one-dimensional dielectric multilayer stack. The hypothesis is that when a block copolymer (BCP) self-assembles into a 50–100+ bilayer lamellar stack, sub-nanometre thermodynamic fluctuations during deposition create a unique depth profile for every tag fabricated. This may generate a complex, multi-peak spectral barcode arising from disorder-induced transmission anomalies in the photonic structure.

No prototype has been fabricated. No spectral measurements have been taken. The concept is at the pre-experimental stage. Its viability depends on answering several critical open questions—foremost among them whether the spectral fingerprint is stable under environmental ageing. Until that question is answered experimentally, the concept remains a hypothesis.

The purpose of this whitepaper is to invite collaborators in PUF hardware security, BCP chemistry, spectral measurement, information theory, and adversarial machine-learning analysis to subject this concept to experimental and cryptographic scrutiny.

2. Background

2.1 The Optical PUF Field

The optical PUF field contains numerous two-dimensional and three-dimensional implementations: colloidal crystal PUFs (Nanoscale, 2022), fluorescent quantum dot PUFs (Liu et al., 2019), holographic PUFs, plasmonic nanoparticle PUFs (Smith et al., 2016), and cholesteric liquid crystal (CLC) PUFs. Most of these encode their entropy in the spatial (x–y) domain and require high-resolution microscopy or imaging for authentication. Some existing approaches—notably CLC PUFs—already use spectral readout rather than imaging, and any comparison with the present proposal must account for these.

2.2 Z-Axis Encoding and Readout

The concept proposed here encodes information in depth—the sequence of internal layer thicknesses within a 1D dielectric stack. The hypothesis is that these internal thicknesses resist non-destructive characterisation by common optical methods (such as simple reflection and transmission spectroscopy from the tag surface), since the spectral output is a complex, non-invertible function of all layer thicknesses simultaneously.

This resistance claim requires formal analysis. Techniques including spectroscopic ellipsometry, cross-sectional transmission electron microscopy (TEM), and focused ion beam (FIB) milling can profile multilayer structures with varying degrees of destructiveness and

spatial resolution. A rigorous security evaluation must enumerate the relevant probe modalities and assess which, if any, could reconstruct the internal profile with sufficient fidelity to clone the tag. This analysis has not been performed and is listed as an open question in Section 4.

2.3 Spectral Feature Sensitivity

The 1D architecture reduces the readout to a single spectral measurement rather than a two-dimensional image. Whether this enables low-cost authentication depends on the spectral resolution required, which is itself an open question.

In low refractive-index-contrast systems (Δn approximately 0.008–0.1), spectral features tend to be narrow (FWHM on the order of 15–40 nm). The sensitivity of these narrow features to sub-nanometre layer-thickness variations is the proposed mechanism for generating tag-unique spectra. **[HYPOTHESIS]** This sensitivity claim requires experimental verification in the specific BCP material system under consideration.

3. The Conceptual Framework

3.1 Prior Art Assessment

A prior-art search was conducted across USPTO, EPO, and WIPO patent databases and academic literature (Google Scholar, Web of Science) during January–February 2026. Search terms included “1D photonic crystal PUF,” “z-axis physically unclonable function,” “multilayer thickness stochastic authentication,” and “spectral PUF dielectric.” No prior work was identified proposing a PUF based on stochastic z-axis thickness variations in a 1D dielectric multilayer stack.

This search was limited to unclassified, English-language sources and independent verification is required. The absence of prior art should not be interpreted as evidence of merit—it may equally indicate that domain experts have considered and discarded the approach for reasons not apparent to the present author. One purpose of this whitepaper is to invite exactly that feedback.

3.2 Encoding Capacity Estimate

The theoretical encoding capacity can be estimated under idealised assumptions. Consider N bilayer pairs, each with a stochastic thickness variation of standard deviation σ of approximately 1.5 nm (consistent with KRICT spin-coated BCP data reporting $\sigma = 1.2$ –2.0 nm). If the measurement system can distinguish thickness differences of δ , then the number of distinguishable states per layer is approximately $S = 6\sigma/\delta$, covering plus or minus three standard deviations.

Under the most optimistic assumption ($\delta = 0.3$ nm, implying sub-nanometre spectral discrimination), this yields approximately 30 states per layer. For 50 layers, the combinatorial space reaches approximately 10^{74} ; for 100 layers, approximately 10^{148} . These numbers are theoretical upper bounds and are likely to be substantially overstated for several reasons.

[HYPOTHESIS] Layer independence assumption. In BCP self-assembly, adjacent lamellar thicknesses are physically correlated through chain connectivity, confinement effects, and substrate interactions. The actual number of independent degrees of freedom may be significantly fewer than N . A correlation-corrected estimate has not been calculated; doing so requires experimental measurement of the inter-layer thickness correlation function, which is a prerequisite for any credible entropy estimate.

[HYPOTHESIS] Resolution assumptions contain an internal contradiction. The $\delta = 0.3$ nm thickness discrimination assumed above implies spectral resolution far exceeding what is available from current chip-scale spectrometers, which typically offer approximately 10 nm spectral resolution (for instance, the Hamamatsu C12666MA micro-spectrometer at 340–850 nm range with approximately 15 nm resolution, or the AMS AS7341 multi-spectral sensor providing discrete-band rather than continuous spectral measurement). There is an approximately 30 $\times$ gap between the assumed measurement capability and what is commercially available at low cost. If the effective δ is constrained by spectrometer resolution rather than by the physics, the number of distinguishable states per layer drops to perhaps 2–5 rather than 30, reducing the encoding capacity by many orders of magnitude.

The appropriate metric for PUF security evaluation is min-entropy per bit, not raw combinatorial capacity. Standard PUF evaluation metrics—inter-PUF Hamming distance distribution, intra-PUF reliability (bit error rate under environmental variation), and min-entropy per bit (Ruhrmair and Holcomb, 2014)—have not been measured and must be determined experimentally before any security claims can be validated. The encoding capacity estimate is included to demonstrate that the proposed mechanism could, in principle, generate sufficient entropy for authentication, but the actual achievable entropy is an open experimental question and may be orders of magnitude lower than these upper bounds.

3.3 Proposed Sub-Applications

Three sub-applications have been identified, all requiring experimental validation. The first is **moiré spectral overlay authentication**: overlaying two slightly mismatched 1D multilayer stacks could create spectral-domain Vernier beating patterns sensitive to alignment and thickness. No prior art was identified for free-space spectral moiré PUFs, though this absence has not been exhaustively verified.

The second is **disorder-induced spectral anomaly authentication**. Disorder-induced transmission anomalies in 1D photonic crystals are known physics (Fan et al., 2002). The specific application of these anomalies to authentication has not been identified in the literature, though the search was not exhaustive.

The third sub-application involves **harnessing BCP self-assembly stochasticity as a PUF entropy source**. A critical caveat applies: BCP process development typically aims to minimise thickness variation for quality control. If manufacturing optimisation succeeds in reducing σ , the PUF entropy decreases. The proposal implicitly requires that stochastic variation be preserved or controlled within a target range during fabrication—a requirement that may conflict with standard manufacturing objectives.

3.4 Projected Security Properties

[HYPOTHESIS] The hypothesised security against cloning and machine-learning attacks rests on two claims, neither of which has been formally proven.

The first claim concerns resistance to physical probing: the internal layer-thickness sequence is hypothesised to resist reconstruction by non-destructive external measurement. As discussed in Section 2, this claim requires formal analysis against specific probe modalities (ellipsometry, TEM, FIB, optical coherence tomography, and others). The claim as stated is not falsifiable without such enumeration, and should be treated as an assumption to be tested rather than an established property.

The second claim concerns resistance to computational modelling: the non-linear relationship between layer thicknesses and spectral output at multiple incidence angles is hypothesised to create a large space of challenge–response pairs that resists machine-learning inversion. Modern generative adversarial networks and neural-network-based inverse-design tools have successfully attacked physically unclonable functions in other domains (Ruhrmair et al., 2010; Ganji et al., 2016), and inverse design of multilayer optical structures is an active research area (Peurifoy et al., 2018). A formal adversarial machine-learning security analysis is required before any security claim can be considered established.

The security argument as currently constructed contains circular elements: security derives from assumed inaccessibility of internal layers, which derives from the 1D architecture, which was chosen in part because it provides security. Independent experimental proof of resistance to layer-profile reconstruction is needed to break this circularity. A complete threat model must also assess thermal, acoustic, mechanical, and other potential side channels through which an attacker might extract information about the internal structure.

3.5 Target Applications

[HYPOTHESIS] The following applications are speculative and conditional on positive outcomes from the stability, readout, and security validation studies described in Section 4. No application should be pursued until spectral fingerprint stability under environmental ageing is confirmed.

Luxury spirits and wine authentication is one candidate, where the PUF label could simultaneously serve as a decorative element (labradorescent visual effect) and an authentication feature. Pharmaceutical supply chain verification represents a second possibility, with potential links to distributed ledger technology for chain-of-custody tracking with offline authentication capability. High-value document authentication (passports, certificates, banknotes) and fine art provenance verification round out the prospective application areas.

These domains were selected because they involve high-value items where the cost of a per-item PUF tag and readout infrastructure might be justified. No market adoption analysis has been performed. Each of these markets has entrenched incumbent authentication technologies (holograms, RFID, DNA taggants) with established supply chains, regulatory acceptance, and user familiarity. The practical barriers to displacing incumbents—including regulatory approval timelines, integration with existing workflows, and end-user training—have not been assessed and would need to be addressed before any application is viable.

4. Discussion and Limitations

4.1 Open Questions Requiring Collaboration

The questions below are listed in order of criticality. The first is the prerequisite gate—a negative answer terminates the concept.

Spectral fingerprint stability. How stable is the spectral fingerprint under environmental ageing (temperature cycling, UV exposure, humidity)? A PUF whose fingerprint drifts over time is useless for authentication. Without demonstrated stability over the required product lifetime (months to years depending on application), no authentication deployment is viable. No accelerated ageing data exists. This is a binary pass/fail gate for the entire concept.

Layer-thickness distribution. What is the measured statistical distribution of layer-thickness variations in BCP self-assembled multilayers, and what is the inter-layer thickness correlation structure? Published data (KRICT, Korea) reports standard deviations of 1.2–2.0 nm for spin-coated films, but this needs verification for alternative deposition methods (slot-die coating, screen printing) at throughputs relevant to manufacturing. Throughput targets have not been defined and would depend on the application—spirits labelling may require tens of thousands of tags per hour, whilst fine-art provenance may require only single-unit fabrication. The correlation structure between adjacent layers is essential for calculating a realistic entropy estimate.

Spectrometer resolution requirements. What spectrometer resolution is required to reliably distinguish individual PUF labels? Can commercially available chip-scale spectrometers—such as the Hamamatsu C12666MA (approximately 15 nm resolution, 340–850 nm) or AMS AS7341 multi-spectral sensor (8 discrete channels)—achieve sufficient discrimination? If sub-nanometre resolution is required, the per-tag authentication cost rises substantially (likely requiring benchtop spectrometers in the £5,000–£50,000 range), restricting viable applications to high-value items with controlled authentication environments.

Adversarial machine-learning resistance. Can adversarial machine-learning attacks reconstruct the z-axis depth profile from captured spectral fingerprints? A formal security analysis using state-of-the-art generative adversarial networks and neural-network inverse-design methods is required. The analysis should also assess non-spectral side channels (thermal, acoustic, mechanical) that might leak structural information.

Enrolment and authentication protocol. What are the practical requirements for an enrolment and authentication protocol? This includes initial fingerprint capture procedure, database architecture and storage requirements, match-threshold selection, false-accept and false-reject rate targets, scalability to large tag populations (10^6+), and protocol resilience to replay attacks. This operational layer is essential to any authentication system and has not been addressed in this document. It represents a substantial engineering effort in its own right.

4.2 Spectral Fingerprint Drift

[HYPOTHESIS] This is the critical risk and is concept-terminating if the outcome is negative.

Temperature cycling causes BCP lamellar d-spacing to change due to polymer thermal expansion. UV exposure photo-oxidises polymer chains, altering refractive index. Moisture absorption can swell polymer layers. A fingerprint captured at enrolment may not match the same tag after extended storage. This is the authentication system's core failure mode and a single point of failure for the entire concept.

If drift exceeds the measurement noise floor, the false-reject rate increases until the entire deployed tag population becomes unauthenticatable. This degradation is not graceful—it crosses a threshold catastrophically at population scale, where all tags manufactured in a given batch fail simultaneously. No accelerated ageing data exists to assess this risk.

4.3 Readout Resolution Gap

There is an approximately 30× gap between the spectral resolution assumed in the encoding capacity estimate ($\delta = 0.3$ nm thickness discrimination) and the resolution available from current chip-scale spectrometers (approximately 10 nm). If the concept requires resolution at the assumed end, authentication would require laboratory-grade equipment, restricting viable deployment to controlled environments (customs inspection, laboratory verification) rather than point-of-sale or consumer use. This does not terminate the concept but substantially narrows its applicability and removes the primary advantage over existing PUF technologies that already work with laboratory readout.

4.4 Undefined Operational Protocol

No enrolment or authentication protocol has been designed. The operational requirements—initial fingerprint capture, database storage, match threshold, false-accept and false-reject rate targets, and scalability to millions of tags—are entirely unaddressed. The hardest engineering challenge in deploying any PUF-based authentication system is typically this operational protocol layer, not the PUF physics itself.

4.5 Interdisciplinary Team Requirement

The concept requires simultaneous expertise in BCP chemistry, photonic crystal physics, spectral measurement engineering, PUF security evaluation, and adversarial machine learning. No single existing team is known to span all of these. The proposal's viability depends on assembling a bespoke interdisciplinary collaboration—a structural fragility. If any one discipline cannot be recruited, the validation programme stalls.

4.6 Process Optimisation Conflict

BCP manufacturing process development typically aims to minimise layer-thickness variation to improve optical quality. If process optimisation succeeds in reducing σ below the threshold needed for sufficient PUF entropy, the authentication concept fails. The PUF application requires controlled disorder—enough variation for uniqueness but not so much that spectral features become unreadable. Whether this operating window exists, and whether it is compatible with manufacturing economics, is unknown.

5. Conclusion

This whitepaper presents a pre-experimental concept for a physically unclonable function that encodes entropy in the z-axis of a 1D dielectric multilayer stack, exploiting the stochastic thickness variations inherent in BCP self-assembly. The concept is at the hypothesis stage. No prototype exists, no measurements have been taken, and several open questions could individually terminate the approach.

The single most critical question is spectral fingerprint stability under environmental ageing. A negative answer renders the concept unviable regardless of progress on any other front. The second critical gap is the approximately 30× mismatch between the spectral resolution assumed in the encoding capacity estimate and the resolution available from commercially viable chip-scale spectrometers.

This document is issued as an open invitation to research collaboration. Critical feedback that identifies fatal flaws early is as valuable as supportive collaboration. All intellectual property generated through collaboration would be jointly held under terms agreed prior to commencement of work.

Contact: aaron@garcia.ltd

6. References

Liu, Y. et al. (2019) Inkjet-printed unclonable quantum dot fluorescent anti-counterfeiting labels with artificial intelligence authentication. *Nature Communications*, 10, 2409.

Fan, S., Suh, W. and Joannopoulos, J.D. (2002) Temporal coupled-mode theory and the presence of non-orthogonal modes in lossless multimode systems. *Physical Review B*, 65(23), 235112.

Cracking enabled unclonability in colloidal crystal patterns authenticated with computer vision (2022). *Nanoscale*. DOI: 10.1039/D2NR01479C.

Ganji, F., Tajik, S. and Seifert, J.-P. (2016) PAC Learning of Arbiter PUFs. *Journal of Cryptographic Engineering*. DOI: 10.1007/s13389-016-0119-4.

Pappu, R. et al. (2002) Physical One-Way Functions. *Science*, 297(5589), 2026–2030.

Peurifoy, J. et al. (2018) Nanophotonic particle simulation and inverse design using artificial neural networks. *Science Advances*, 4(9), eaar4206.

Ruhrmair, U. and Holcomb, D.E. (2014) PUFs at a Glance. *DATE Conference*.

Ruhrmair, U., Sehnke, F., Solter, J., Dror, G., Devadas, S. and Schmidhuber, J. (2010) Modeling attacks on physical unclonable functions. *Proceedings of the ACM Conference on Computer and Communications Security*, 237-249.

Smith, A.F., Patton, P. and Skrabalak, S.E. (2016) Plasmonic nanoparticles as a physically unclonable function for responsive anti-counterfeit nanofingerprints. *Advanced Functional Materials*, 26(8), 1315-1321.

Appendix A: Document Metadata

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Contact	aaron@garcia.ltd
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Appendix B: Glossary

Term	Definition
BCP	Block copolymer; a polymer comprising two or more distinct monomer blocks that can self-assemble into ordered nanostructures
CLC	Cholesteric liquid crystal; a liquid crystal phase with a helical structure that produces spectral-selective reflection
FIB	Focused ion beam; a technique using a focused beam of ions for imaging, milling, and cross-sectioning of materials at the nanometre scale
FWHM	Full width at half maximum; a measure of spectral feature bandwidth
PUF	Physically unclonable function; a physical structure that produces a unique, device-specific response when challenged, used for authentication
TEM	Transmission electron microscopy; an imaging technique using electrons transmitted through an ultra-thin specimen
Δn	Refractive index contrast; the difference in refractive index between alternating layers in the multilayer stack
δ	Measurement resolution; the minimum thickness difference distinguishable by the readout system
σ	Standard deviation of layer-thickness variation in the BCP self-assembly process
d-spacing	The characteristic repeat distance (period) of the lamellar structure in a block copolymer film
Lamellar	A layered or plate-like morphology formed by block copolymer self-assembly, producing alternating thin layers
Min-entropy	A conservative measure of the unpredictability of a random variable, used as the security-relevant entropy metric for PUF evaluation
Hamming distance	A metric quantifying the difference between two binary strings; used in PUF evaluation to measure inter-device and intra-device variability
Spectroscopic ellipsometry	An optical technique for characterising thin film thickness and optical properties by measuring changes in polarisation state of reflected light
Vernier beating	An interference pattern produced when two periodic structures of slightly different periods are overlaid

Appendix C: Methodology and Transparency Statement

Author Context and Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The analytical framework reflects professional history, focusing on practical application. The author is an independent researcher with no university or institutional affiliation. The framework presented here reflects practical, cross-domain experience but lacks the domain-specific depth that PUF security researchers, BCP chemists, or photonic crystal physicists would bring—which is why this document is framed as a collaboration invitation.

Digital Methodology and Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools—specifically Claude, Google Gemini, and ChatGPT—to assist in the production of this paper. These tools were employed as force multipliers for data synthesis and editorial accessibility. As a writer with dyslexia, I use these models as assistive technology to refine grammar and streamline sentence structure.

Verification and Integrity

Whilst AI assisted in processing data, the intellectual oversight is entirely human. I performed manual validation of all citations and accept full responsibility for the accuracy and originality of the final output. Several references are marked as representative citations requiring full bibliographic verification.

Appendix D: Raw Author Notes

The concept for a z-axis PUF originated from Project Labradorite, an investigation into synthetic labradorescence using BCP multilayer thin films. The observation that self-assembly inherently produces stochastic layer-thickness variations—normally treated as a manufacturing defect to be minimised—prompted the question of whether this randomness could be repurposed as a security feature.

The intellectual leap was recognising that a “defect” in one context (optical quality manufacturing) could be a “feature” in another (authentication). This reframing is the core original contribution. Everything else—the encoding capacity estimates, the security arguments, the application mapping—follows from that single insight and awaits experimental validation.

Internal project documents informing this work include the Project Labradorite Commercial Applications deep research report, the Strategic Evaluation and Commercial Roadmap, the Critical Evaluation and Strategic Enhancement external review, and the Improvements Enhancement Brief—all produced in early 2026. These documents are not publicly accessible; all material claims are self-contained within this whitepaper.

Appendix E: Change Log

Version	Date	Description
1.0	25 February 2026	Publication-ready document prepared from author source material. Structured as a Scientific/Conceptual collaboration whitepaper. All content reviewed for accuracy, consistency, and appropriate qualification of claims. Evidence classification tags applied throughout.

End of document.